

Technical Specification of 5KV Digital Insulation Tester

- **Scope:** 5KV Digital Insulation Resistance Tester to check quality of insulation in electrical equipments like Distribution/ Power Transformers, Current Transformers, Cables and other electrical insulations. The kit shall be AC (mains) and battery operated, rugged & portable with easy to operate functions.
- **Climatic Conditions:** The High Voltage Test Kit should be suitable for satisfactory operation under the following environmental conditions.
 - Operating Temperature: 0°C to 50°C
 - Storage Temperature: -20°C to 70°C
 - Relative Humidity: <90% RH, Non Condensing
- **General Specification:**
 - The kit should display Test Voltage, Insulation Resistance, Leakage Current and Capacitance after each test.
 - Kit should have latest diagnostic modes like IR, PI, DAR, DD, Step Voltage and BDV.
 - The kit should have self-calibration checkup at every startup.
 - The kit should auto discharge the device under test at the end of each test.
 - The kit should have Bluetooth connectivity.
 - The kit should have optional feature to connect with Android application, if subscribed.
 - The kit should be battery operated & lightweight for field operations.
 - The kit should be mains & battery operated, in case of battery failure still it should work on mains supply.
 - It should have Li-Ion battery with back up of approx. 4 hours continuous testing and approx. charging time 3 hours. In case of Battery drainage, the instrument should run on AC Voltage Power source.
 - The Kit should have replaceable battery.
 - The Kit should have auto-power off feature.
 - The kit & accessories should comply international standards.
 - Instrument should be supplied with windows software for operating instrument from PC and generating the report.
 - The manufacturer should have service center in India
 - The following particular shall be clearly marked in a visible position
 - Manufacturer Name/Logo
 - Model Number
 - Serial Number
- **Technical Specification:**
 - Diagnostic Modes: IR, PI, DAR, DD, Step Voltage & Breakdown Voltage.
 - Test Voltage: DC Volt- (250VDC to 5,000VDC), Programmable in steps of 50V
Fixed steps of 250V, 500V, 1KV, 2.5KV, 5KV
Accuracy: ±5%
DC Volt- 0- 750v
 - Short circuit current: 5mA
 - Insulation Resistance Range: Digital display: 1KΩ to 50TΩ
 - Accuracy: Up to 1 TΩ @ ±5%
1TΩ to 50TΩ @ ±10%.
 - Leakage Current Measurement: upto 6mA
Accuracy: ±5%
 - Capacitance Measurement: 10nF to 50 μF
Accuracy: ±(10% of reading + 5 nF)

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- AC/DC Voltage Measurement: up to 600 V AC/DC
Accuracy: $\pm (5\% \text{ of rdg} + 5 \text{ digits})$
- Timer: Three Programmable Timers
Timer Range: Programmable from 1 Sec to 99 Min. 59sec.
- Power Supply:
Mains Supply: 100 - 230V AC, 50Hz $\pm 10\%$
Battery Supply: Li-Ion Battery
- Battery backup: Minimum 4 hours of continuous testing.

Charging time: 3 hours or less. Fast charging preferred.

- Guard Accuracy: 2% error guarding <300 K Ω leakage with 100 M Ω load
- Interference rejection: 1mA rms per 250V up to a max of 4mA.
- Display: LCD/ LED with backlit
- PC interface: USB/RS232
- Memory: Internal memory for 1000 Test Results.
- Software: Dedicated software for data transfer & report generation
- Device should have strong anti interference ability.

➤ **Conformity Standard:**

Safety In accordance with EN 61010-1
EMC In accordance with EN 61326-1
Type Test In accordance with IS 11994
Ingress Protection IP 65
CE & CAT III 600V / CAT IV 600V
Safety of Accessories with IEC61010-31

➤ **Weight:** <4Kg (Excluding accessories)

➤ **Indications & Annunciations**

Audio visual indication for HV presence
LED indication for HV presence
Battery status indication, Battery charging indication
Audible discharge annunciation
Mains / Battery Indication

• **Display details**

Real-time display of Insulation Resistance
After end of test scrolling display of test results including resistance, leakage current, time & Capacitance

➤ **Protection:** The kit should have thermal & Overload protection which can prevent damages to Internal circuit of the kit.

➤ **Accessories:**

- 3 X 3mtr Heavy Duty Lead set
- 3 X 15 mtr Heavy Duty Lead set
- Power Cord
- Communication cable: RS232 male to female serial cable 9 pin/ USB 2.0 A/B cable 1.5 mtr long


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- Instruction Manual
- Calibration Certificate
- Carrying Bag
- Mains Protected fuse
- Bluetooth Software
- Desktop Application

- **Documents:** Following documents/certificates shall be furnished by the bidder along with offer
- Guaranteed Technical Particulars (GTP).
 - The calibration certificate should be provided by the supplier with the equipment.
 - Quality Assurance Plan (QAP).

Eligibility Criteria for Participation [Qualification Requirement (QR)]

1. Participation in the Tender

- Original manufacturers of Instruments in India will be eligible in the tender.
- For any non-indigenous Manufacturer having either manufacturing or assembling unit in India in collaboration with any registered Indian firm, are also eligible to quote.
- Distributor or Dealer or Agent or Channel Partner will be eligible for submission of quotation.
- Bidder will be OEM/OES in nature.
- Instruments should be calibrated from any NABL accredited Laboratory. Calibration certificate should bear NABL logo in each page. Calibration results shall comply with the specifications.
- Bids will be accepted only after meeting the technical requirement as per specification and on site-demonstration

2. Experience: Experience in the field of manufacturing of Testing Instruments for last 5 (five) years. (Documents to be submitted in support of claims)

3. Warranty:

- a) Manufacturer shall invariably give unconditional Warranty for minimum 05 years from the date of demonstration of the materials on delivery. The Warranty shall include calibration of all the Instruments including transportation during Warranty period. The terms and condition of the Warranty shall be attached along with the Technical Bid. The bidder shall quote separately for extended Warranty, if they wish. Regarding non indigenous Manufacturers having either manufacturing or assembling unit in India in collaboration with any registered Indian Company, the Warranty should be provided by the Parent Company.
- b) All the Instruments supplied against this specification shall be Warranted for a defect liability of at least 60 months from the date of receipt at store duly passed acceptance testing at after delivery.
- c) Any Engineering error, omission, wrong provision, Instrument failure etc., the Instrument shall be attended by the bidder free of cost during the Warranty period
- d) For any defect in the kit during the Warranty period, repairing has to be completed within 15 days. Otherwise substitute set of same type has to be provided within 15 days from the date of reporting. The substitute set will be handed back to supplier on receipt of original set [duly repaired].
- e) If any defect observed during Warranty period, it is bidder's responsibility to collect kit from site /consignee to attend the same.
- f) On each failure of Test set, the supplier shall submit detailed report of failure analysis to Distribution Testing Department, WBSEDCL
- g) Necessary Life time licensed copy of interfacing and analysis software with test kit should be supplied free of cost with test Instrument.

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4. **Demonstration:** During Technical scrutiny of the bid, the purchaser will ask to demonstrate the offered Instrument and requisite accessories at their own cost at any specified location given by the Purchaser. The Bidder shall have to demonstrate the test Instrument within 15 days of intimation from the purchaser. It is the responsibility of the bidder to bring all the Instruments with necessary software and accessories required for Demonstration of the features available in the test set as per Technical Specification at date and time communicated to them in advance. The bidders are also required to provide the operating manual /required literature and sent competent Engineers at that. The purchaser will evaluate the suitability of product of their application. The purchaser's decision will be final for acceptance/ non-acceptance of product.
5. **Performance Certificate:** Bidder shall submit at least two Performance certificate from any power utility/ Govt. Department for Instruments supplied during last 5 (five) years from date of Tender to be submitted with the Offer. The date of performance certificate should be at least after one year from the date of delivery. Documentary evidence for the supply & acceptance of quantity by the Consignee Authority to be submitted with the offer.
6. **Maintenance and support:** Bidders must give necessary service and spare part support for the entire life time of the Testing Instruments offered after completion of the Warranty period as and when required.
7. **On-site installation, commissioning and demonstration:**
The supplier shall be responsible for the **on-site installation, commissioning, and demonstration** of the electrical instruments at the delivery location specified by the purchaser.
The scope of the demonstration shall include:
- Unpacking and proper installation as per manufacturer's specifications.
 - Electrical safety checks and power connection under supervision of the Purchaser's technical team.
 - Demonstration of all operating functions and settings.
 - Verification of performance as per Technical Specifications in the P.O.
 - Basic training to end-users/maintenance personnel, including do's and don'ts, safety procedures, and routine maintenance tips.
- The supplier must obtain a **Demonstration Completion Certificate**, signed by the purchaser's authorized representative.
- No additional charges shall be claimed for demonstration, travel, or manpower.
8. **Packing and Forwarding:** The equipment along with accessories shall be properly packed to withstand the handling during transportation and outdoor storage during transit. The easily damageable materials shall be carefully packed. Each consignment shall be accompanied by a detailed packing list.
9. **Type Test Certificate:** Submission of Type test certificate preferably from CPRI [Central Power Research Institute, Ministry of Power, Govt. of India, Bangalore] or ERDA [Electrical Research and Development Association, Mumbai] Laboratory or any other NABL accredited Laboratory/ International Standard Laboratory for the offered Instrument is mandatory.


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GTP of the Digital Insulation Tester: 5 kV

(To be submitted by the Bidder)

Sl no	Description	
1.	Name of the Manufacturer	
2.	Address	
3	Reference Standard	
4	Size and Weight of the Instrument	
5	Address of Service Centre in India	
6	Operating Condition	
	(i) Temperature	
	(ii) Relative Humidity	
7	Rated Output Voltage	
8	Mains Supply	
9	Resistance Range	
10	Standard	
11	Accuracy	
12	Accessories	
	Ingress Protection	
13	Length of Power Cord to be supplied	
14	Length of Measuring cable	
15	Short Circuit Current	
16	Programmable Test Voltage	
17	Display Type (Analog/ Digital)	
18	Timer (0 sec to 99 min)	
19	Battery Life	
20	Interface	
21	Safety	
22	Auto Power off	


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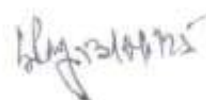

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